entanglement with Photon B, the disruption caused by my measurement is imprinted on the distant Photon C.

And that's fantastically interesting. Through my measurement, we are able to learn how Photon A's spin is related to Photon B's, but with the prickly problem that both photons were disrupted by my meddling. Through entanglement, however, Photon C is tied in to my measurement —even though it's thousands of miles away—and this allows us to isolate the effect of the disruption and thereby have access to information ordinarily lost in the measurement process. If I now call Nicholas with the result of my measurement, he will learn how the spins of Photons A and B are related after the disruption, and, via Photon C, he will have access to the impact of the disruption itself. This allows Nicholas to use Photon C to, roughly speaking, subtract out the disruption caused by my measurement and thus skirt the obstacle to duplicating Photon A. In fact, as Bennett and collaborators show in detail, by at most a simple manipulation of Photon C's spin (based on my phone call informing him how Photon A is spinning relative to Photon B) Nicholas will ensure that Photon C, as far as its spin goes, exactly replicates the quantum state of Photon A *prior to my measurement.* Moreover, although spin is only one characteristic of a photon, other features of Photon A's quantum state (such as the probability that it has one energy or another) can be replicated similarly. Thus, by using this procedure, we could teleport Photon A from New York to London.[3]

As you can see, quantum teleportation involves two stages, each of which conveys critical and complementary information. First, we undertake a joint measurement on the photon we want to teleport with one member of an entangled pair of photons. The disruption associated with the measurement is imprinted on the distant partner of the entangled pair through the weirdness of quantum nonlocality. That's Stage 1, the distinctly quantum part of the teleportation process. In Stage 2, the result of the measurement itself is communicated to the distant reception location by more standard means (telephone, fax, e-mail . . .) in what might be called the classical part of the teleportation process. In combination, Stage 1 and Stage 2 allow the exact quantum state of the photon we want to teleport to be reproduced by a straightforward operation (such as a rotation by a certain amount about particular axes) on the distant member of the entangled pair.